Name:

Quiz 1 Solutions

Instructions. The only tools you are permitted to use are pencils, pens, erasers, a sheet of handwritten notes, and your mind. No electronic devices! You have 1 hour. Be sure to sign the Honor Code statement on the last page. Good luck!

Problem 1 (True-false, 7 points)**.** You will get 1 point for a correct answer and 0 points for blank and incorrect answers.[1] No explanations are required.

(1)	The ODE $y' + x^2 y = \sin(x)$ is linear.	**T**	F
(2)	Every autonomous first order ODE is separable.	**T**	F
(3)	Every function y satisfying $y' = x - \cos(y)$ must be increasing for all $x > 1$.	**T**	F
(4)	If f is any function such that $f' = 1 - f$, then $\lim_{x\to\infty} f(x) = 1$.	**T**	F
(5)	The initial value problem $y' = x^2 + y^2$ and $y(0) = 1$ has a unique solution.	**T**	F
(6)	The initial value problem $(y - x)y' = 0$ and $y(0) = 0$ has a unique solution.	T	**F**
(7)	Consider the initial value problem $y' = y$ and $y(0) = -1$, whose solution is $y = -e^x$. If we run Euler's method with step size 0.2 to approximate $y(1)$, we end up underestimating the true value of $y(1)$.	T	**F**

[1] During quiz revisions, you'll have the chance to meet with me one-on-one and convince me that you understand and can fully explain *up to 2 of the true-false questions that you left blank* to get full credit on those questions.

Problem 2 (2 points)**.** Solve the following initial value problems.

(a) $y' = -2xy^2$ and $y(0) = 1$.

Solution. Note that $y \equiv 0$ is a solution to the ODE, but it does not satisfy the initial condition. Other solutions to the ODE can be found by separating variables.

$$\begin{aligned} -\frac{dy}{y^2} &= 2x\,dx \\ \frac{1}{y} &= x^2 + C \\ y &= \frac{1}{x^2 + C} \end{aligned}$$

Plugging in $x = 0$ and $y = 1$, we find $C = 1$, so the solution is

$$y = \frac{1}{x^2 + 1}.$$

(b) $y' + 2y = 3e^x$ and $y(0) = 5$.

Solution. This equation is linear, so we can use the method of integrating factors. An integrating factor is

$$r(x) = e^{\int 2\,dx} = e^{2x}$$

which means that

$$(e^{2x}y)' = e^{2x}y' + 2e^x y = 3e^{2x}e^x = 3e^{3x}.$$

Antidifferentiating with respect to x, we get

$$e^{2x}y = e^{3x} + C$$

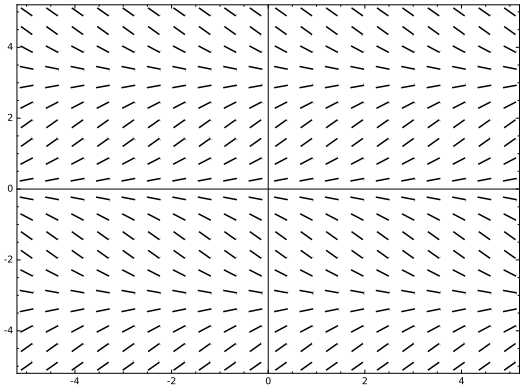
which means that

$$y = e^x + Ce^{-2x}.$$

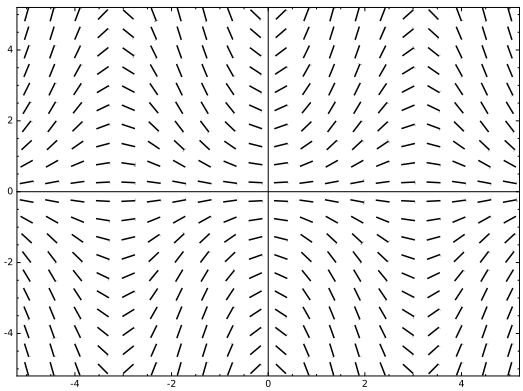
Plugging in $x = 0$ and $y = 5$, we get $C = 4$, so the solution is

$$y = e^x + 4e^{-2x}.$$

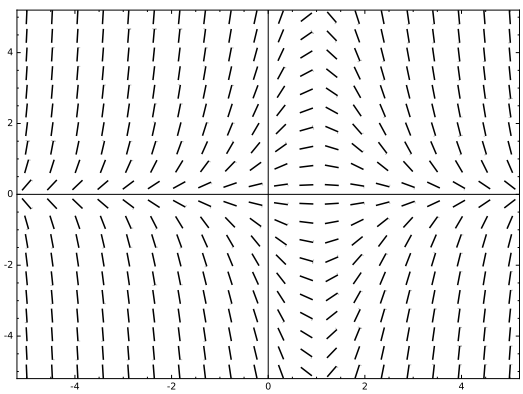
Problem 3 (3 points)**.** Match each ODE to its slope field.



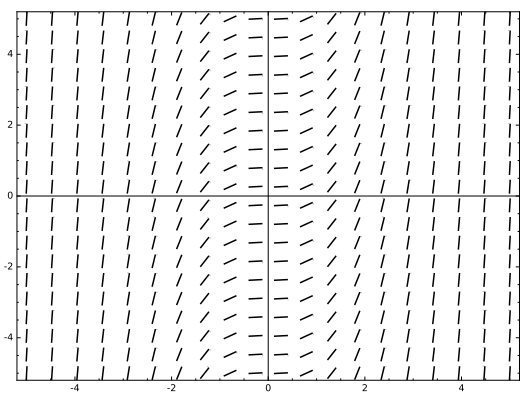
(a) $y' = x^2$



(b) $y' = \sin(y)$



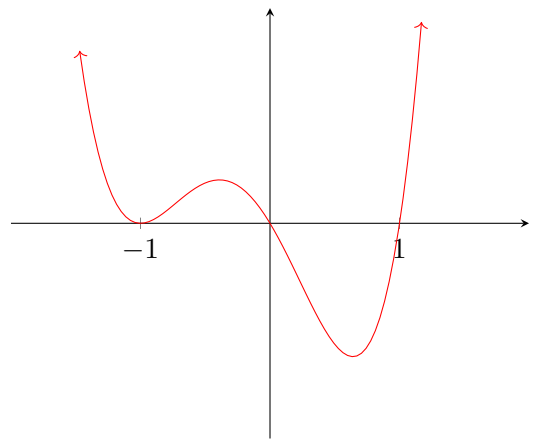
(c) $y' = y\sin(x)$



(d) $y' = (1 - x)y$

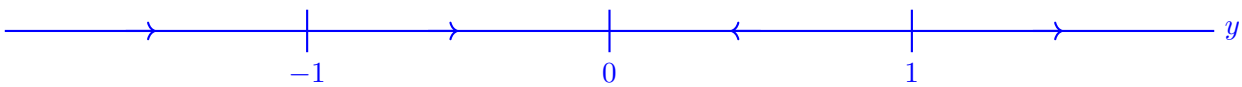
Solution. In the order that the slope fields are displayed, the corresponding ODEs are (b), (c), (d), (a).

Problem 4 (3 points)**.** Consider the autonomous ODE $y' = f(y)$, where f is the polynomial function whose graph is depicted below.



(a) Draw a phase diagram. Which, if any, of the critical points are stable?

Solution.



There are 3 critical points, at $y = -1, 0, 1$.

Only $y = 0$ is stable (ie, a sink). -1 is semistable (ie, a node) and 1 is unstable (ie, a source).

(b) If y is a solution to the ODE satisfying $y' = f(y)$ the initial condition $y(0) = -0.5$, what is $\lim_{x\to\infty} y(x)$?

Solution. $\lim_{x\to\infty} y(x) = 0$.

Problem 5 (3 points)**.** A tank starts off containing 10 L of pure water. At time t (measured in minutes), salt enters the tank at a rate of $10t$ mg/min and immediately gets uniformly dissolved in the water. The contents of the tank drain out of the tank at a rate of 2 L/min. Let $a(t)$ denote the amount of salt, in mg, in the tank at time t.

(a) Write down an ODE that a satisfies. It should involve no other variables except a and t.

Solution. At time t, there are $10t$ mg of salt entering the tank, and the amount leaving is $2 \cdot a/V$ where V is volume of water in the tank. We know that $V(t) = 10 - 2t$. Thus

$$\frac{da}{dt} = 10t - \frac{2a}{10 - 2t} = 10t - \frac{a}{5 - t}.$$

(b) Indicate what method you would use to solve the ODE (ie, "separation of variables" or "integrating factors"). You do *not* need to actually solve it.

Solution. This is a linear first order ODE, so it can be solved using the method of integrating factors.

Problem 6 (Extra credit! 2 points, no partial credit)**.** Let f be the continuous function

$$f(y) = \begin{cases} y^{1/3} & \text{if } y \geq 0 \\ -y^{1/3} & \text{if } y < 0. \end{cases}$$

Sketch the phase diagram for $y' = f(y)$. Is it possible to find two functions g_1 and g_2, both of which are continuously differentiable on all real numbers and both of which satisfy the initial value problem

$$y' = f(y) \quad y(-3/2) = -1,$$

but where

$$\lim_{x\to\infty} g_1(x) = 0 \quad \text{and} \quad \lim_{x\to\infty} g_2(x) = \infty?$$

If this is possible, find such functions g_1 and g_2 and show that they have the required properties. If this is impossible, explain why.

Solution. I'll tell you that it is possible to find such functions, but I'm intentionally omitting a detailed solution so that you can think about it ☺

Honor code. If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

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